

## Basic Java Exam - Standardized

1. Write a Java program to swap two integer variables 'a' and 'b' without using a third variable.

Demonstrate this with a = 5 and b = 10.

2. Write a Java program using a for loop to print the first 10 positive integers and their squares in the format:

1 -> 1

2 -> 4

...

10 -> 100

3. Write a Java program that uses a for loop and conditionals to print all even numbers from 1 to 20 inclusive. For each even number, print the message: 'This is an even number.'

4. Declare and initialize an array of five integers in Java. Write a program to compute and print the average of these integers.

5. Given the array: `int[] numbers = {1,2,3,4,5,6,7,8,9,10};` write a Java program to calculate the sum of all elements using a for loop.

6. Modify the program in Question 5 to skip the number 5 during summation.